PROBLEM 8 | Problems T2_CrystStruct.pdf

Crystalline Structures & Miller Indices

Planar Packing Fraction on FCC Planes

Original statement · Problems T2_CrystStruct.pdf, p. 2

> *8. Calculate the fraction of area occupied by atoms for the* (111), (200), (220), (222), (400) *and* (420) *planes, in the FCC structure.* *(Solution:* $(111) : \pi/(2\sqrt{3})$*;* $(200) : \pi/4$*;* $(220) : \pi\sqrt{2}/8$*;* $(222) : 0$*;* $(400) : 0$*;* $(420) : \pi/(4\sqrt{5})$*)*

PHASE 1: HYPOTHESIS & PARAMETERS

Lattice Model:

- Structure: Face-Centred Cubic (FCC).
- Fundamental contact relation along $\langle 110 \rangle$:

$$a = 2\sqrt{2}R \implies R = \frac{a}{2\sqrt{2}} = \frac{a\sqrt{2}}{4}$$

- Cross-sectional area of each spherical atom of radius R:

$$A_{\text{atom}} = \pi R^2$$

PHASE 2: FUNDAMENTAL PHYSICS & FORMULA SELECTION

The **Fraction of Area Occupied by Atoms (Planar Packing Fraction, PPF)** on a crystallographic plane $(h\,k\,l)$ is defined as the ratio between the total area of the circular sections of the atoms whose centres fall exactly on that plane and the total geometric area of the plane inside the unit cell [Session 4 Slide 9]:

$$\text{PPF}_{(hkl)} = \frac{N_{\text{atoms}} \cdot (\pi R^2)}{A_{(hkl)}}$$

- If a plane passes between atomic planes and **does not cut the centre of any atom**, the number of atoms contained is zero and $\mathrm{PPF} = 0$.
